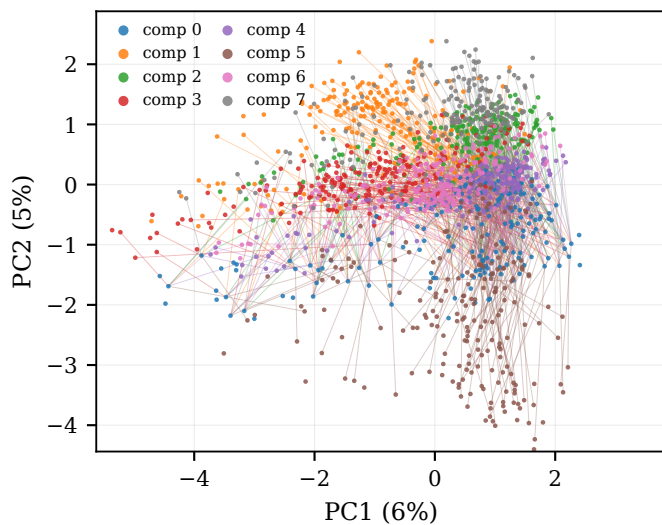
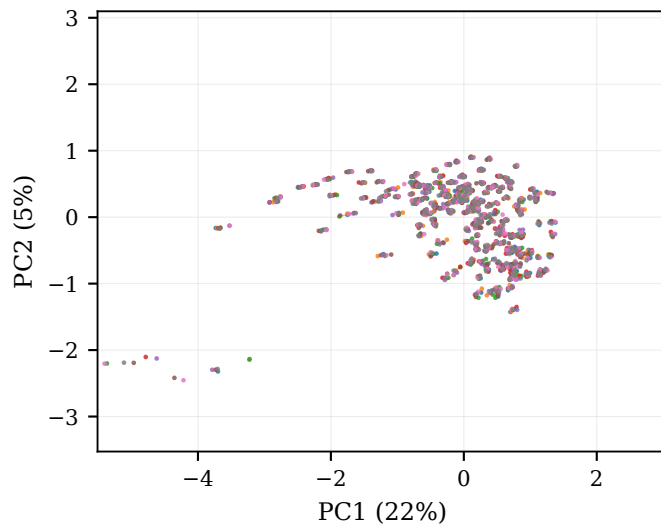


Same 4096 tokens, forwarded under all 8 compartments, layer-4 residual in one shared PCA basis
each line joins a token in compartment 0 to its twin in compartment j

Default init
(no alignment pressure)
 $\cos = 0.132$ twin dist = $1.37 \times$ cloud radius



Init-copy
(embeddings + head shared at init)
 $\cos = 0.981$ twin dist = $0.20 \times$ cloud radius



InfoNCE
(explicit layer-4 alignment)
 $\cos = 0.963$ twin dist = $0.25 \times$ cloud radius

